

FractionalMaxPool2d#

<https://pytorch.org/docs/stable/generated/torch.nn.FractionalMaxPool2d.html#>

Description:

Applies a 2D fractional max pooling over an input signal composed of several input planes. Fractional MaxPooling is described in detail in the paper Fractional MaxPooling by Ben Graham The max-pooling operation is applied in $kH \times kW \times kH \times kW$ regions by a stochastic step size determined by the target output size. The number of output features is equal to the number of input planes. Exactly one of `output_size` or `output_ratio` must be defined.

Function Signature

```
class torch.nn.FractionalMaxPool2d(kernel_size,  
output_size=None, output_ratio=None, return_indices=False,  
_random_samples=None)[source]#
```

Parameters

Shape:

Code Examples

```
>>> # pool of square window of size=3, and target output size
13x12
>>> m = nn.FractionalMaxPool2d(3, output_size=(13, 12))
>>> # pool of square window and target output size being half of
input image size
>>> m = nn.FractionalMaxPool2d(3, output_ratio=(0.5, 0.5))
>>> input = torch.randn(20, 16, 50, 32)
>>> output = m(input)
```

Notes & Warnings

NOTE Exactly one of `output_size` or `output_ratio` must be defined.

Detailed Documentation

Documentation

Applies a 2D fractional max pooling over an input signal composed of several input planes.

Fractional MaxPooling is described in detail in the paper Fractional MaxPooling by Ben Graham

The max-pooling operation is applied in $kH \times kW$ regions by a stochastic step size determined by the target output size. The number of output features is equal to the number of input planes.

Exactly one of `output_size` or `output_ratio` must be defined.

`kernel_size` (Union[int, tuple[int, int]]) – the size of the window to take a max over. Can be a single number k (for a square kernel of $k \times k$) or a tuple (kh, kw)

`output_size` (Union[int, tuple[int, int]]) – the target output size of the image of the form $oH \times oW$. Can be a tuple (oH, oW) or a single number oH for a square image $oH \times oH$. Note that we must have $kH + oH - 1 \leq H_{in}$ and $kW + oW - 1 \leq W_{in}$

`output_ratio` (Union[float, tuple[float, float]]) – If one wants to have an output size as a ratio of the input size, this option can be given. This has to be a number or tuple in the range $(0, 1)$. Note that we must have $kH + (output_ratio_H * H_{in}) - 1 \leq H_{in}$ and $kW + (output_ratio_W * W_{in}) - 1 \leq W_{in}$

`return_indices` (bool) – if True, will return the indices along with the outputs. Useful to pass to `nn.MaxUnpool2d()`. Default: False

Input: (N, C, H_{in}, W_{in}) or (N, C, H_{in}, W_{in}) or (C, H_{in}, W_{in})

Output: (N, C, H_{out}, W_{out}) or (C, H_{out}, W_{out}) , where $(H_{out}, W_{out}) = output_size(H_{out}, W_{out})$ or $(H_{out}, W_{out}) = output_ratio \times (H_{in}, W_{in})$

